

Spatio-Temporal Joint Filtering for Transfer Learning in Brain-Computer Interface of Motion Imagery

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Abstract—This paper presents a novel joint spatio-temporal filtering algorithm for transfer learning (TL) in motion imagery (MI) tasks, aiming to address the variability in subjects, trials, or tasks. The algorithm combines spatial filtering using the common spatial pattern (CSP) algorithm with temporal filtering applied to EEG signals of each channel. By jointly optimizing the coefficients of the spatial and temporal filters, the algorithm enhances the performance of CSP and improves the alignment of covariance matrices. Experimental results on various public datasets demonstrate that the proposed joint spatio-temporal filtering algorithm effectively improves the classification accuracy of TL for MI tasks in brain-computer interfaces (BCI).

Index Terms—Brain-computer interfaces (BCI), common spatial pattern (CSP), electroencephalograph (EEG), joint spatio-temporal filtering, transfer learning.

I. INTRODUCTION

Brain-computer interface (BCI) systems [1]-[4] establish a direct communication pathway between individuals and their environment by utilizing brain signals. Among the various modalities used in BCI, electroencephalography (EEG) [5] stands out as a non-invasive technique that records brain activity from electrodes placed on the scalp. EEG-based BCI has witnessed significant advancements, offering great potential for various applications. EEG-based BCI relies on capturing and analyzing electrical signals generated by the brain to infer the user's intentions or cognitive states. Several established paradigms are utilized, including motor imagery (MI) [7], event-related potentials (ERPs) [8], and steady-state visual evoked potentials (SSVEP) [6]. In MI tasks, individuals imagine performing specific movements, generating MI-based EEG signals. Accurately classifying MI-EEG signals is crucial for the success of MI-based BCI [9].

The common spatial pattern (CSP) algorithm [10], introduced in [23] and widely used in classifying cognitive states with multi-channel EEG signals, is a popular spatial filtering technique. It identifies spatial filters that extract relevant features for classification. CSP transforms EEG signals into a new space to emphasize significant variance differences between different tasks. The extracted features can then be fed into machine learning algorithms like LDA or SVM for classification [11]. In BCI systems, CSP relies on features obtained from covariance matrices of EEG signals, filtered

in the spatial or temporal domain. Recent studies have highlighted the direct manipulation of covariance matrices for MI classification, including the minimum distance to Riemannian mean (MDRM) classifier [12]-[13]. These advancements contribute to improved BCI accuracy and expand the potential applications of MI-based classification.

BCI systems encounter challenges in dealing with the varying neural responses across different subjects and even within the same subject for different stimuli, times, or locations. Acquiring subject-specific labeled training data during BCI calibration can be expensive and time-consuming. To tackle these issues, transfer learning (TL) techniques [17]-[20] have been adopted in BCI systems. TL leverages data from a source domain to enhance learning performance in a target domain. TL approaches, as proposed in [14] and [15], aim to construct a common feature space to improve the estimation performance of CSP filters. Samek *et al.* [16] developed a TL-based method for transferring non-stationary information in data, reducing subject-specific variations and achieving promising results in MI-BCI. Zanini *et al.* [21]-[22] introduced the Riemannian alignment (RA) framework, which aligns EEG covariance matrices using resting-state information from different subjects. RA is based on the MDRM classifier. Additionally, He and Wu [24] extended RA to Euclidean alignment (EA) in Euclidean space, enabling the use of any Euclidean space classifier. By leveraging TL techniques, BCI systems can overcome subject variability and reduce the reliance on extensive subject-specific training data. TL approaches, including common feature space construction, non-stationary information transfer, and covariance matrix alignment, contribute to improved performance and broader applicability in MI-based BCI systems.

This paper examines the application of joint spatio-temporal filtering in TL tasks, emphasizing the effectiveness of filtering techniques in addressing spectral and channel response differences caused by variations in subjects or experimental conditions. The study investigates the impact of joint spatio-temporal filtering on the alignment performance of covariance matrices in domain adaptation. The structure of the paper is as follows. Section II introduces the proposed joint spatio-temporal filtering algorithm, providing an overview of its prin-

ciples and methodology. Section III presents the experimental results, offering a comprehensive analysis of the algorithm's performance. Finally, Section IV concludes the paper by summarizing the main findings, discussing the implications, and suggesting future research directions.

II. PROPOSED SPATIO-TEMPORAL JOINT FILTERING ALGORITHM

In traditional CSP, feature extraction is performed solely using spatial filters. However, EEG signals are known to be highly nonstationary in nature. To enhance classification accuracy, it is beneficial to incorporate temporal filtering techniques as well. In this subsection, we will introduce an efficient algorithm for designing both spatial and temporal filters based on the CSP framework.

Assume that the source domain contains n_S labeled samples $\{\mathbf{X}_{S,i}, y_{S,i}\}_{i=1}^{n_S}$, where $\mathbf{X}_{S,i} \in \mathbb{R}^{C \times T}$ is the i th sample data matrix and $y_{S,i} \in \{1, \dots, l\}$ represents its corresponding label. Let $\mathbf{y}_S = [y_{S,1}, \dots, y_{S,n_S}] \in \mathbb{R}^{n_S \times 1}$ be the label vector of the source domain. Furthermore, the target domain has n_T unmarked $\{\mathbf{X}_{T,i}\}_{i=1}^{n_T}$, where $\mathbf{X}_{T,i} \in \mathbb{R}^{C \times T}$. Here, C and T are the numbers of EEG channels and temporal samples, respectively. The traditional CSP problem can be formulated as follows

$$\max_{\mathbf{w} \in \mathbb{R}^c} \frac{\mathbf{w}^T \mathbf{C}_1 \mathbf{w}}{\mathbf{w}^T \mathbf{C}_2 \mathbf{w}} \quad (1)$$

where $\mathbf{w} = [w_1, w_2, \dots, w_c]^T$ represents the coefficient vector of the spatial filter, $\mathbf{C}_1$ and $\mathbf{C}_2$ are the sample covariance matrices of signals of different categories, which are obtained by

$$\mathbf{C}_i = \frac{1}{|\mathcal{C}_i|} \sum_{k \in \mathcal{C}_i} \frac{\mathbf{X}_k \mathbf{X}_k^T}{\text{Tr}\{\mathbf{X}_k \mathbf{X}_k^T\}}, \quad i = 1, 2, \quad (2)$$

where $\mathcal{C}_i$ denotes the index set of data blocks associated with the i th category. To solve ((1)), the generalized eigenvalue decomposition of $\mathbf{C}_1$ and $\mathbf{C}_1 + \mathbf{C}_2$ is employed to achieve

$$\mathbf{C}_1 \mathbf{u}_j = \lambda_j (\mathbf{C}_1 + \mathbf{C}_2) \mathbf{u}_j \quad (3)$$

where λ_j represents the j th generalized eigenvalue, $\mathbf{u}_j$ denotes the corresponding eigenvector. Without the loss of generality, eigenvalues $\{\lambda_j\}$ are sorted in a non-ascending order, that is, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_c$. Then, the optimal solution to ((1)) is the leading eigenvector $\mathbf{u}_1$.

A. Spatio-Temporal Joint Filter

The proposed model in this paper, referred to as the spatio-temporal joint filter model, builds upon the foundation of the CSP framework. The primary objective of this model is to enhance the discrimination between two distinct classes by maximizing the variance ratio between the filtered signals. To train the spatio-temporal joint filter, sample data from the source domain, denoted as $\{\mathbf{X}_{S,i}\}$, is employed. The formulation of the spatio-temporal joint filter model can be summarized as follows:

$$\max_{\mathbf{w} \in \mathbb{R}^c, \mathbf{h} \in \mathbb{R}^L} \frac{\mathbf{w}^T \mathbf{C}_1^h \mathbf{w}}{\mathbf{w}^T \mathbf{C}_2^h \mathbf{w}} \quad (4)$$

where $\mathbf{h} = [h_1, \dots, h_L]^T$ represents the temporal filter coefficient vector with order L . The spatio-temporal joint filtering model constructs a pair of spatio-temporal filters $\mathbf{w}$ and $\mathbf{h}$ for samples of each class. Let $\mathbf{x}_{S,i,k}^T$ be the k th channel of EEG signals in $\mathbf{X}_{S,i}$. To perform the temporal filtering, $\mathbf{x}_{S,i,k}^T$ is first segmented, yielding

$$\tilde{\mathbf{X}}_{S,i,k} = \begin{bmatrix} x_{S,i,k}(0) & x_{S,i,k}(1) & \dots & x_{S,i,k}(L-1) \\ x_{S,i,k}(1) & x_{S,i,k}(2) & \dots & x_{S,i,k}(L) \\ \vdots & \ddots & \ddots & \vdots \\ x_{S,i,k}(T-L) & x_{S,i,k}(T-L+1) & \dots & x_{S,i,k}(T-1) \end{bmatrix} \quad (5)$$

where $x_{S,i,k}(t)$ denotes the EEG signal of the k th channel of source domain EEG signals in $\mathbf{X}_{S,i}$ obtained at a sampling instant. Using $\tilde{\mathbf{X}}_{S,i,k}$ for $k = 1, \dots, C$, the temporal filtering process of $\mathbf{X}_{S,i}$ can be expressed as $\tilde{\mathbf{X}}_{S,i,k} \mathbf{h}$, which is then used to construct the sample covariance matrices.

B. Alternating Update Optimization Algorithm

In this paper, we employ an alternative optimization approach to tackle the joint design problem outlined in (4). Our proposed technique involves iterative iterations, where the algorithm focuses on fixing either the spatial or temporal filter and subsequently updating the coefficient vector of the other filter. This process leads to an optimization problem that bears resemblance to (1) and can be efficiently solved using a closed-form solution. By adopting this strategy, we are able to significantly enhance the computational efficiency of the algorithm.

Given the spatial filter coefficient vector $\mathbf{w}$, we can update the temporal filter by directly solving (4) using the generalized Eigenvalue Decomposition (EVD) technique. Once we have successfully determined $\mathbf{w}$, (4) can be expressed as follows

$$\max_{\mathbf{h} \in \mathbb{R}^L} \frac{\mathbf{h}^T \mathbf{C}_1^w \mathbf{h}}{\mathbf{h}^T \mathbf{C}_2^w \mathbf{h}} \quad (6)$$

To perform the spatial filtering, $\mathbf{x}_{S,i,k}^T$ is further segmented, yielding

$$\tilde{\mathbf{X}}_{S,i,k} = \begin{bmatrix} x_{S,i,k}(0) & x_{S,i,k}(1) & \dots & x_{S,i,k}(C-1) \\ x_{S,i,k}(1) & x_{S,i,k}(2) & \dots & x_{S,i,k}(C) \\ \vdots & \ddots & \ddots & \vdots \\ x_{S,i,k}(T-C) & x_{S,i,k}(T-C+1) & \dots & x_{S,i,k}(T-1) \end{bmatrix} \quad (7)$$

Using $\tilde{\mathbf{X}}_{S,i,k}$, the spatial filtering process of $\mathbf{X}_{S,i}$ can be expressed as $\tilde{\mathbf{X}}_{S,i,k} \mathbf{w}$, which is then used to construct the sample covariance matrices.

The optimization problem stated in (7) can indeed be effectively solved using the generalized Eigenvalue Decomposition (EVD) method. The aforementioned steps are iteratively repeated until the updates of the filter coefficients $\mathbf{w}$ and $\mathbf{h}$ become negligible.

C. Domain adaptation

In TL, domain adaptation (DA)[18] is a widely used approach that focuses on aligning the data distributions between

the source and target domains. The goal is to bridge the gap between these domains so that knowledge learned from the source domain can be effectively transferred to the target domain. As one of popular DA techniques, EA aligns the feature distributions of the source and target domains by minimizing their Euclidean distance, such that features from both the source and target domains can be mapped to a shared feature space. Unlike EA, RA is based on the covariance matrices of the source and target domains, rather than the EEG trails.

In the context of TL, we employed a spatio-temporal joint filter to train on the source domain data, resulting in temporal and spatial filter coefficients $\mathbf{h}$ and $\mathbf{w}$. These coefficients were then utilized to process the data, generating processed source domain data $\{\tilde{\mathbf{X}}_{S,i}\}$ and target domain data $\{\tilde{\mathbf{X}}_{T,i}\}$. To incorporate this method into the framework of transfer learning, we integrated it with a DA approach, following these steps:

- 1) Feature alignment: We employed the DA method to align the features of the processed source domain data and the unprocessed target domain data. This alignment procedure aimed to reduce the distributional differences between the two domains, enabling effective transfer learning.
- 2) Classifier training: Using the aligned features, we trained a classifier (such as LDA, logistic regression, or neural networks) on labeled data from the target task. By leveraging the aligned features and the corresponding target domain labels, the classifier learned the relevant patterns and relationships specific to the target domain.

The Riemannian distance between SPD matrices $\mathbf{C}_1$ and $\mathbf{C}_2$ is first defined as

$$\delta(\mathbf{C}_1, \mathbf{C}_2) = \|\log(\mathbf{C}_1^{-1}\mathbf{C}_2)\|_F, \quad (8)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The Riemannian mean $\mathbf{M}_R$ can be computed by

$$\mathbf{M}_R = \arg \min_{\mathbf{C}} \frac{1}{n} \sum_{i=1}^n \delta^2(\mathbf{C}, \tilde{\mathbf{C}}_{S,i}) + \frac{1}{m} \sum_{j=1}^m \delta^2(\mathbf{C}, \tilde{\mathbf{C}}_{T,j}), \quad (9)$$

The Euclidean mean is:

$$\mathbf{M}_E = \frac{1}{n+m} \left(\sum_{i=1}^n \tilde{\mathbf{C}}_{S,i} + \sum_{j=1}^m \tilde{\mathbf{C}}_{T,j} \right). \quad (10)$$

where $\tilde{\mathbf{C}}_{S,i}$ and $\tilde{\mathbf{C}}_{T,j}$ denote the covariance matrices of filtered EEG data blocks $\tilde{\mathbf{X}}_{S,i}$ and $\tilde{\mathbf{X}}_{T,j}$ belonging, respectively to source and target domains. Therefore, we perform EA on $\tilde{\mathbf{X}}_{S,i}$ and $\tilde{\mathbf{X}}_{T,i}$:

$$\mathbf{X}'_{T,i} = \mathbf{M}_E^{-\frac{1}{2}} \tilde{\mathbf{X}}_{T,i} \quad (11a)$$

$$\mathbf{X}'_{S,i} = \mathbf{M}_E^{-\frac{1}{2}} \tilde{\mathbf{X}}_{S,i} \quad (11b)$$

RA first computes the covariance matrices of some resting (or non-target) trials, where the subjects did not perform any task

(or did not perform the target task), and then calculates the Riemannian mean $\mathbf{M}_R$ of these matrices through the following transformation. This matrix is used as a reference matrix to reduce variations across sessions or subjects. Using this reference matrix, $\tilde{\mathbf{C}}_{S,i}$ and $\tilde{\mathbf{C}}_{T,j}$ are then subjected to RA as follows:

$$\mathbf{C}'_{S,i} = \mathbf{M}_R^{-\frac{1}{2}} \tilde{\mathbf{C}}_{S,i} \mathbf{M}_R^{-\frac{1}{2}} \quad (12a)$$

$$\mathbf{C}'_{T,i} = \mathbf{M}_R^{-\frac{1}{2}} \tilde{\mathbf{C}}_{T,i} \mathbf{M}_R^{-\frac{1}{2}} \quad (12b)$$

where $\tilde{\mathbf{C}}_{S,i}$ and $\tilde{\mathbf{C}}_{T,i}$ denote the covariance matrix of $\tilde{\mathbf{X}}_{S,i}$ and $\tilde{\mathbf{X}}_{T,i}$ respectively.

By combining the spatio-temporal joint filtering and the DA method, we integrated this approach into the framework of transfer learning. This integration enables the transfer of learned knowledge from the source domain to the target domain, enhancing the classifier's performance on the target domain tasks. The joint utilization of these techniques addresses the domain shift by aligning the features and leveraging the learned representations to improve the classifier's generalization and adaptability to the target domain.

III. EXPERIMENTS

A. Datasets

Our experiments utilize two public datasets:

- Dataset 1: This dataset consists of EEG data obtained from seven healthy participants who took part in BCI Competition IV. The dataset includes recordings of Motor Imagery (MI) from two classes, specifically chosen from left hand, right hand, and foot categories. EEG signals were recorded from 59 channels at a sampling rate of 100 Hz. The dataset is divided into three phases: calibration, evaluation, and feature extraction. For our study, we focus on the calibration data, which contains complete label information. In the calibration phase, each participant performed 100 trials for each class.
- Dataset IIIa: This dataset is sourced from BCI Competition III and contains EEG data collected from three healthy participants. Each participant was instructed to perform four different MI tasks involving the left hand, right hand, foot, and tongue. EEG signals were recorded using 60 channels at a sampling rate of 250 Hz. For our study, we selected two MI classes, specifically the left hand and right hand. Each class consisted of 60 trials.

Preprocessing steps were applied to the EEG signals to remove muscle artifacts and DC drift. A bandpass finite impulse response (FIR) filter of order 50 with a passband of 8 to 30 Hz was used. The experimental data was then extracted from the EEG signals recorded between 0.5 and 3.5 seconds after the cue onset. To further refine the data, a temporal filter with an order of 100 (denoted as L) was applied.

In the conducted experiments, the test set consisted of trials from one subject, while the training set included trials from all other participants. The experimental results were reported as the average classification accuracy. All experiments were

performed using MATLAB 2018b on a system equipped with an Intel(R) Core(TM) i5-7200U CPU @ 2.50GHz processor.

B. Baseline Algorithms

In our research, we performed a comparative analysis of classification accuracy between our novel algorithm and conventional approaches using CSP filtering combined with EA/RA applied to TL algorithms. The classifiers employed in our experiments included MDRM and LDA. Based on the combination of CSP filtering techniques and the alignment of covariance matrices employed, the baseline algorithms are categorized as follows:

- Alignment without CSP filtering: EA-MDRM, EA-LDA, RA-MDRM, RA-LDA.
- Alignment with classical CSP filtering: CSP-EA-LDA, CSP-RA-LDA.
- Alignment with joint filtering: Proposed-EA-LDA, Proposed-RA-LDA.

C. Data Visualization Using t -SNE

To visualize how our proposed method reduces inter-individual variability, we used t -Stochastic Neighbor Embedding(t -SNE) [25], a nonlinear dimensionality reduction technique. t -SNE maps each point in high-dimensional space to a position in 2D or 3D space, while optimizing pairwise distances in the reduced space relative to those in the original manifold. In our experiments, we aimed to represent each covariance matrix as a point in a two-dimensional space for visualizing and comparing the differences in distributions before and after the experiments.

In each trial, we selected one subject's trials as the test set and combined trials from all remaining subjects as the training set. Fig.1a and 1b show the t -SNE visualizations of EEG data from the top two participants in Dataset 1. The red dots represent the test trials, while the blue dots represent the training trials. Each row contains two images. The image on the left represents the experiment before spatio-temporal joint filtering, while the image on the right represents the experiment after spatio-temporal joint filtering. The visualizations of EEG data from the top two participants in Dataset IIIa are shown in Fig.2a and 2b.

From Fig. 1 and 2, we can observe that before spatio-temporal joint filtering, the training trials (blue dots) may be scattered far away from the test trials (red dots). As a result, applying a classifier designed on the training trials directly to the test trials may not yield good performance. However, after spatio-temporal joint filtering, the training and test trials overlap, reducing the differences between the two.

In conclusion, t -SNE visualization was used to demonstrate how the spatio-temporal joint filtering method reduces inter-individual variability in BCI classification. The visualizations show that spatio-temporal joint filtering can effectively reduce the differences between training and test trials, improving the generalization performance of the classifier.

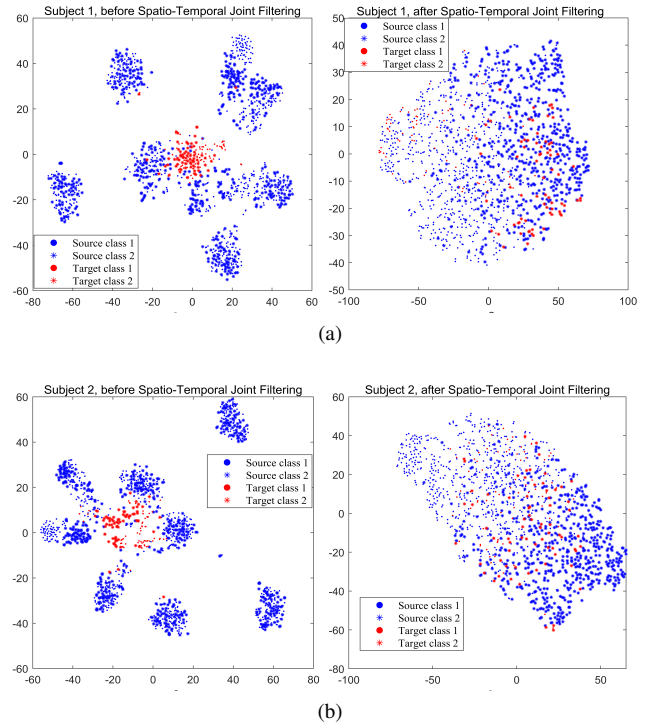


Fig. 1. t -SNE imaging before and after spatio-temporal joint filtering of the first two subjects in Dataset 1.(a) Subject 1; (b) Subject 2. Red dots: trials from test subjects; Blue dots: trials from training subjects.

D. Classification Accuracies

TABLE II
CLASSIFICATION ACCURACY (%) OF DATASET IIIa

subject	1	2	3	Avg
EA-MDRM	50.00	68.33	31.67	50.00
EA-LDA	56.67	70.00	33.33	53.33
RA-MDRM	56.67	70.00	33.33	53.33
RA-LDA	66.67	50.00	51.67	56.11
CSP-EA-LDA	88.33	90.00	91.67	90.00
CSP-RA-LDA	81.67	90.00	91.67	87.78
Proposed-EA-LDA	100.00	93.33	100.00	97.78
Proposed-RA-LDA	100.00	98.33	100.00	99.44

Tables I and II show the classification accuracies on two MI datasets. Compared to the baseline algorithms, our proposed algorithms achieve the best performance in most of experiments.

Based on the experimental results, our proposed algorithm demonstrates superior performance in both Dataset 1 and Dataset IIIa.

In Dataset 1, our algorithm achieves the best performance in six out of seven subjects, outperforming both traditional CSP-based and CSP-free algorithms. This indicates the effectiveness of our proposed method in improving classification accuracy. Furthermore, within the context of TL, the RA algorithm exhibits higher classification accuracy compared to the EA algorithm. This finding suggests that combining our proposed algorithm with the RA algorithm can yield even

TABLE I
CLASSIFICATION ACCURACY (%) OF DATASET I

Subject	1	2	3	4	5	6	7	Avg.
EA-MDRM	80.00	51.50	51.00	60.50	72.00	83.50	52.50	64.43
EA-LDA	73.50	64.50	70.50	56.00	77.50	60.00	82.00	69.14
RA-MDRM	77.00	71.00	64.50	58.00	81.50	83.00	78.00	73.29
RA-LDA	75.00	62.50	71.50	56.00	81.00	58.50	85.00	69.93
CSP-EA-LDA	64.50	58.00	89.50	92.00	93.50	91.00	65.00	79.07
CSP-RA-LDA	76.00	66.50	91.50	92.00	96.00	91.00	70.50	83.36
Proposed-EA-LDA	89.50	89.00	96.00	96.00	98.50	91.50	96.00	93.79
Proposed-RA-LDA	92.50	92.00	96.00	96.50	98.50	93.50	96.50	95.07

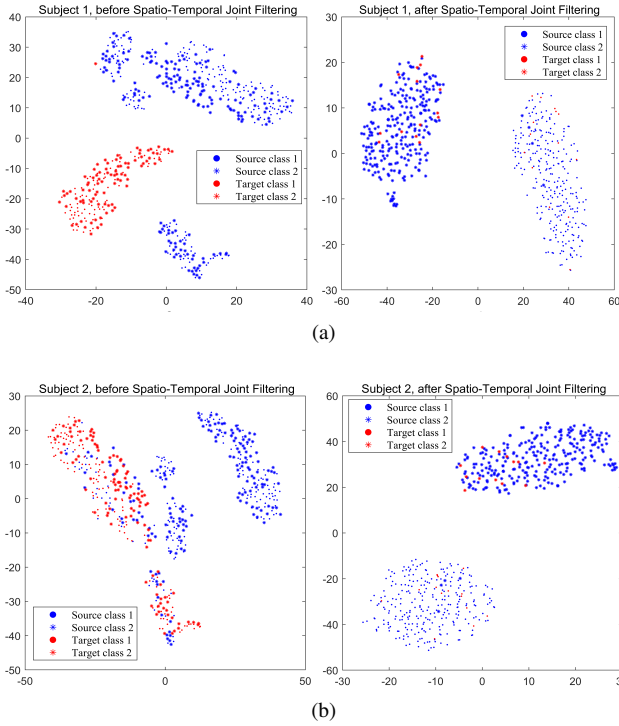


Fig. 2. t -SNE imaging before and after spatio-temporal joint filtering of the first two subjects in Dataset IIIa. (a) Subject 1; (b) Subject 2. Red dots: trials from test subjects; Blue dots: trials from training subjects.

better classification performance, highlighting the potential benefits of leveraging both algorithms together.

For Dataset IIIa, our proposed method consistently achieves the highest classification performance across all three subjects, with particularly notable results for participants 1 and 3. The algorithms based on our proposed method consistently outperform those without CSP and those utilizing traditional CSP. These results indicate that our proposed algorithm, when combined with the RA algorithm and LDA classifier, yields superior classification performance in Dataset IIIa.

In summary, these results demonstrate that our proposed algorithm outperforms traditional CSP methods and yields improved learning outcomes when applied in the context of TL, particularly when combined with the RA algorithm and LDA classifier. These findings highlight the effectiveness and potential of our approach in enhancing classification accuracy and supporting the application of transfer learning in EEG

data analysis. These findings underscore the efficacy and potential of our method in improving classification accuracy and supporting the utilization of TL in EEG data analysis applications.

IV. CONCLUSIONS

Transfer learning in BCI research, based on EEG data, is gaining increasing popularity to address cross-subject and cross-session variability. However, improper experimental arrangements across different subjects can lead to negative transfer due to individual differences. This paper introduces a novel spatio-temporal filter design algorithm that optimizes the filter sparsity of filter coefficients. The spatiotemporal filters are updated using an alternating optimization scheme, and the obtained problem is effectively solved through eigenvalue decomposition while preserving the form of CSP. To mitigate differences between EEG experiments, DA techniques are employed in the spatio-temporal joint filtering process, reducing the gap between the source and target domains, enabling effective transfer of knowledge learned from the source domain to the target domain. Experimental results obtained from publicly available datasets of BCI Competitions III and IV in the context of EEG-based TL scenarios demonstrate the effectiveness of the proposed algorithm.

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